

Summary Report: Scottish SARS-CoV-2 Genomic Cluster Analyses

7 May 2026

1. Purpose

This report summarises the Scottish SARS-CoV-2 genomic cluster analyses intended for three distinct thesis chapters. It outlines the focus of each chapter, the main results, key figures and tables, and a short request for advice on how best to shape the work into a manuscript.

2. Summary

The three analyses form a coherent thesis arc. Part 1 is the strongest analytical chapter and examines deprivation, surveillance, and genomic cluster structure. Part 2 is a descriptive vaccination chapter, showing how cluster vaccination profiles changed through the rollout and across variant waves. Part 3 is a policy-context chapter, describing how cluster structure aligned with Scottish restriction periods while carefully avoiding causal claims because of strong wave and calendar-time confounding.

Across all three parts, the main insight is that genomic clusters reflect a mixture of transmission, local social structure, variant wave, vaccination rollout, surveillance intensity, and policy context. Deprivation matters, but not as a simple pattern of larger or more geographically dispersed clusters in more deprived areas. The clearest deprivation-related signals are in cluster composition and demographic mixing.

3. Chapter Overview

Thesis part	Main focus	Core contribution	Manuscript potential
Part 1	Deprivation, surveillance, and cluster structure	Shows that apparent cluster size and spread are strongly shaped by surveillance and epidemic context, while deprivation signals are clearest in demographic mixing.	Strongest standalone manuscript candidate.
Part 2	Vaccination profiles within clusters	Describes how clusters shifted from unvaccinated to mixed to predominantly vaccinated across rollout and variant waves.	Useful as a descriptive chapter or supporting context.
Part 3	Policy periods and cluster structure	Places cluster dynamics on the Scottish policy timeline and uses targeted ITS analyses at selected transitions.	Best framed as contextual or secondary unless paired carefully with Part 1.

4. Analysis Population

Measure	Value
QC filter	nextclade_qc == "good"
Primary clustering resolution	Leiden 0.3
Unique sequences	281,320
Sequence rows used	789,347
Total inferred clusters	193,160

Measure	Value
Cluster rows used in main Part 1/3 models	193,112
Non-singleton clusters	84,067 in Part 1/3; 84,080 in Part 2 summaries
Sliding time windows	134
Raw Pango lineages	788
Modelled lineage levels after pooling	183

5. Part 1: Socioeconomic Deprivation, Surveillance, and Cluster Structure

5.1. Focus

Part 1 asks whether clusters linked to more deprived areas were larger, more geographically dispersed, or more socially and demographically mixed. Outcomes include cluster size, number of distinct datazones, and observed-minus-expected mixing by SIMD, age, sex, and joint SIMD-age-sex profile.

5.2. Key Results

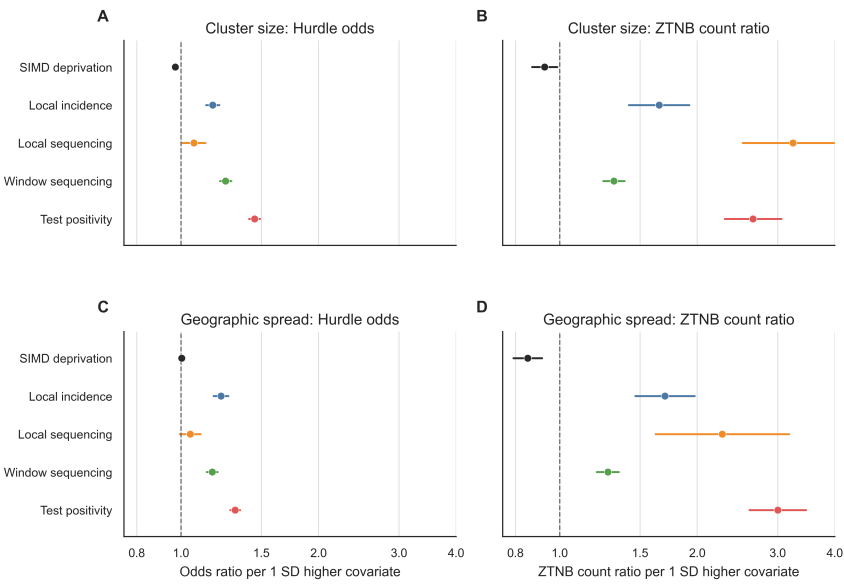
Result	Estimate	Interpretation
SIMD deprivation and non-singleton cluster odds	OR 0.971, 95% CI 0.960 to 0.983	More deprived clusters were slightly less likely to exceed singleton size.
SIMD deprivation and positive cluster size	Count ratio 0.926, 95% CI 0.869 to 0.987	Among non-singleton clusters, higher deprivation was associated with smaller clusters.
SIMD deprivation and multi-datazone spread	OR 1.004, 95% CI 0.992 to 1.016	No clear association with crossing the one-datazone threshold.
SIMD deprivation and positive geographic spread	Count ratio 0.851, 95% CI 0.792 to 0.915	Higher deprivation was associated with lower geographic spread among clusters already spanning multiple datazones.
SIMD-quintile excess mixing	+0.31 percentage points, 95% CI -0.18 to 0.80	No clear evidence of increased SIMD-quintile mixing.
Age excess mixing	+1.66 percentage points, 95% CI 1.29 to 2.03	More deprived clusters showed greater age mixing.
Sex excess mixing	-0.78 percentage points, 95% CI -1.16 to -0.39	More deprived clusters showed lower sex mixing.
Joint SIMD-age-sex excess mixing	+0.48 percentage points, 95% CI 0.29 to 0.67	More deprived clusters showed greater joint demographic-profile mixing.

5.3. Main Insight

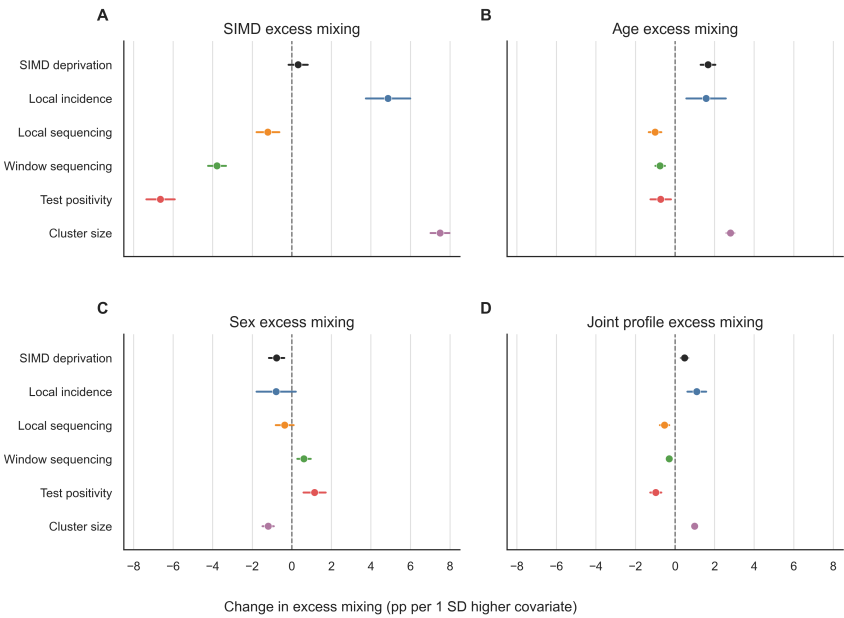
The main models do not support a simple hypothesis that deprivation produces larger or more geographically dispersed genomic clusters. Surveillance and epidemic context, including local incidence, sequencing intensity, window-level sequencing proportion, and test positivity, were more consistent predictors of apparent cluster scale. Deprivation-related signals were clearest in demographic mixing and varied across SIMD domains.

5.4. Key Figures

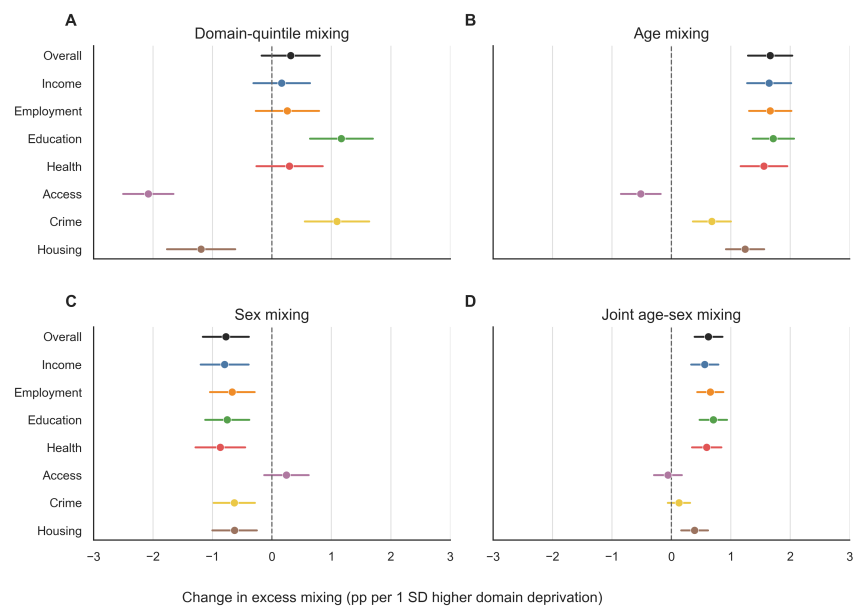
Figure	What it shows
Part 1 Figure 1	Main hurdle and zero-truncated negative-binomial results for cluster size and geographic spread.
Part 1 Figure 2	Adjusted associations with SIMD, age, sex, and joint demographic excess mixing.
Part 1 Figure 3	How different SIMD domains relate to deprivation-domain and demographic mixing.
Part 1 Figure 4	Wave-specific deprivation associations for cluster size and geographic spread.



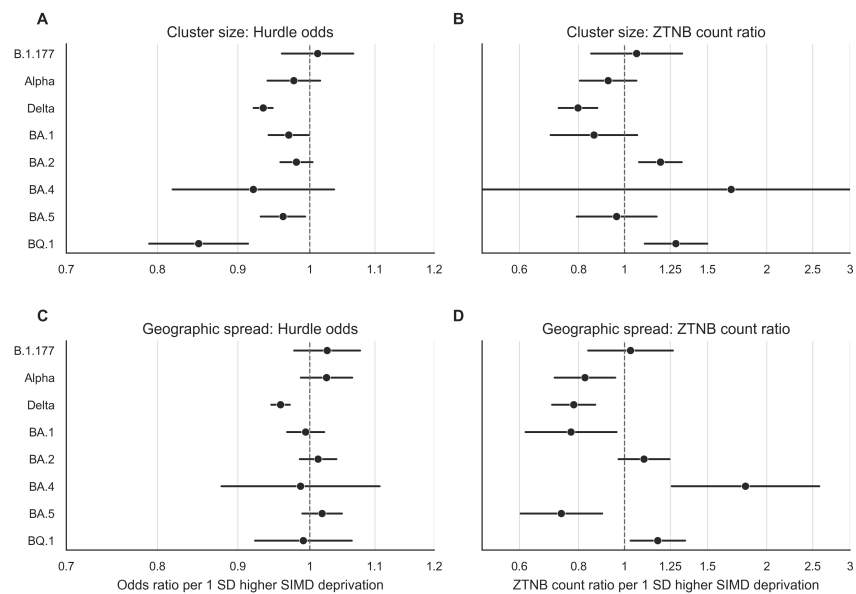
Part 1 Figure 1: Main cluster outcomes



Part 1 Figure 2: Main cluster mixing



Part 1 Figure 3: SIMD-domain mixing



Part 1 Figure 4: Wave-specific cluster outcomes

5.5. Key Tables

Table	File
Dataset descriptives	main_dataset_descriptives.csv
Main hurdle count model results	main_hurdle_count_model_results.csv
Main mixing model results	main_mixing_model_results.csv
SIMD-domain demographic mixing results	main_simd_domain_demographic_mixing_model_results.csv
Wave-specific hurdle count results	main_wave_specific_hurdle_count_model_results.csv

6. Part 2: Vaccination Profiles Within Genomic Clusters

6.1. Focus

Part 2 characterises vaccination status within non-singleton genomic clusters across epidemic waves, cluster size, geographic-dispersion categories, and SIMD quintiles. This is descriptive rather than causal and should not be interpreted as vaccine-effectiveness analysis.

6.2. Key Results

Wave	None vaccinated	Mixed vaccination	All vaccinated
B.1.177	89.5%	10.0%	0.5%
Alpha	41.5%	56.2%	2.3%
Delta	10.3%	65.3%	24.4%
BA.1	4.4%	49.9%	45.7%
BA.2	2.6%	41.0%	56.4%
BA.5	1.8%	27.1%	71.0%
BQ.1	0.6%	20.5%	78.9%
XBB	0.0%	19.4%	80.6%

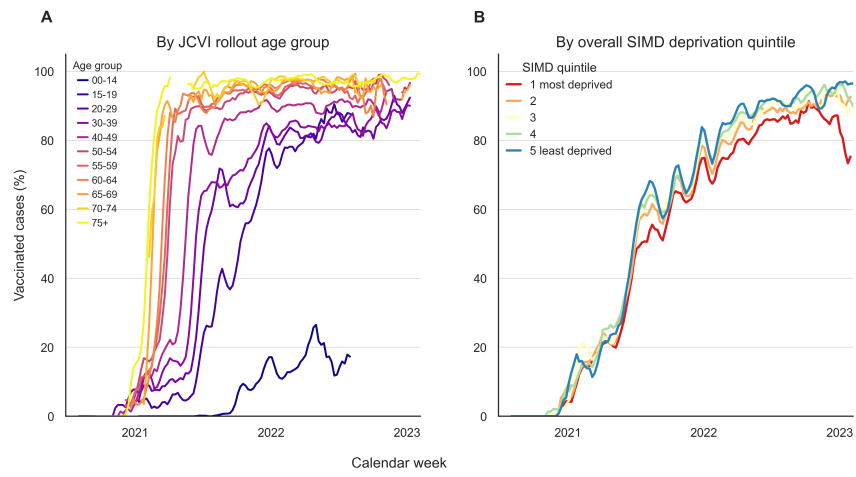
Vaccination-status mixing result	Pattern
Mixed vaccination-status clusters	Peaked in Delta, when 49.1% of non-singleton clusters were more mixed than expected.
Homogeneous vaccination-status clusters	Became dominant from BA.1 onwards, reaching 66.3% in BA.5.
Booster coverage by SIMD	Modest positive Q5-Q1 gradients in most post-Delta waves, indicating higher booster coverage in less deprived clusters.
Dose recency	Median days since vaccination increased from 13 days in B.1.177 to 218 days in XBB.
Large clusters	More likely to be homogeneous by SIMD, consistent with geographic concentration of large transmission events.

6.3. Main Insight

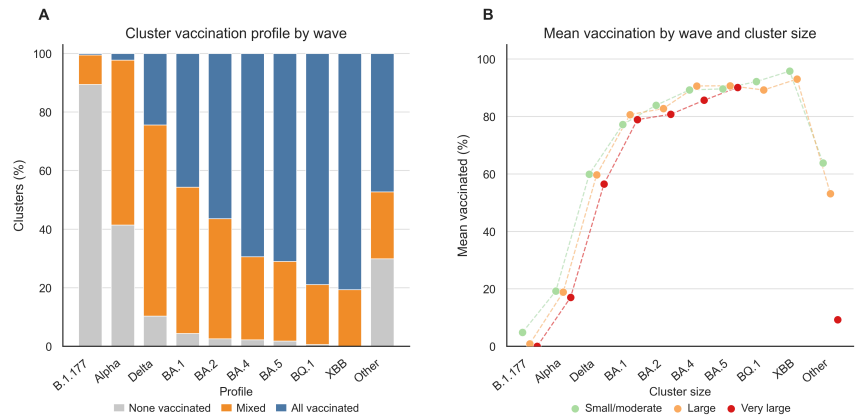
Cluster vaccination profiles largely tracked the national rollout. Mixed vaccination-status clusters were most common when vaccine coverage was changing rapidly across the population. Later clusters were predominantly all-vaccinated because vaccination had become the background condition among sequenced cases, not because vaccinated cases formed exceptional clusters.

6.4. Key Figures

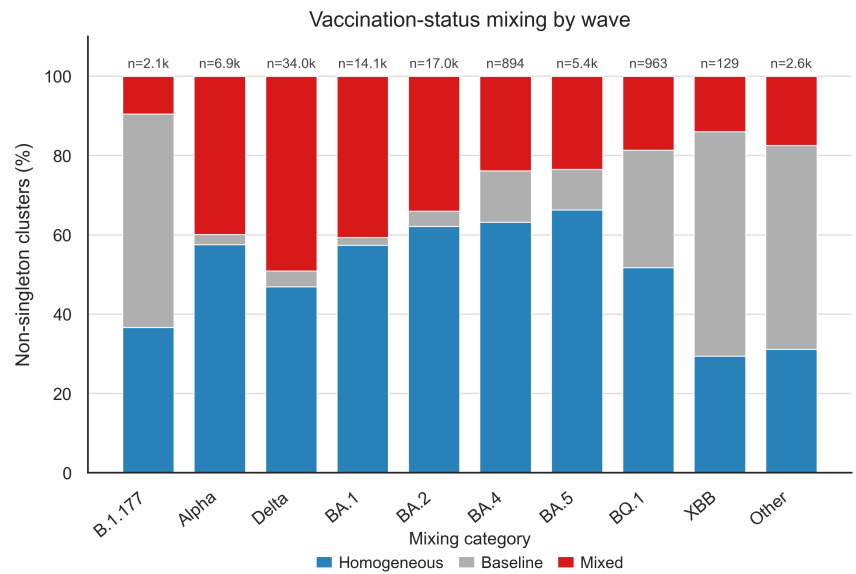
Figure	What it shows
Part 2 Figure 1	Weekly proportion of sequenced cases who were vaccinated, by rollout age group and SIMD quintile.
Part 2 Figure 2	Shift from unvaccinated to mixed and all-vaccinated clusters across waves and size categories.
Part 2 Figure 3	Vaccination-status mixing categories by wave, relative to same-lineage/window expectations.
Part 2 Figure 6	Booster coverage and days since last vaccination by wave and SIMD quintile.



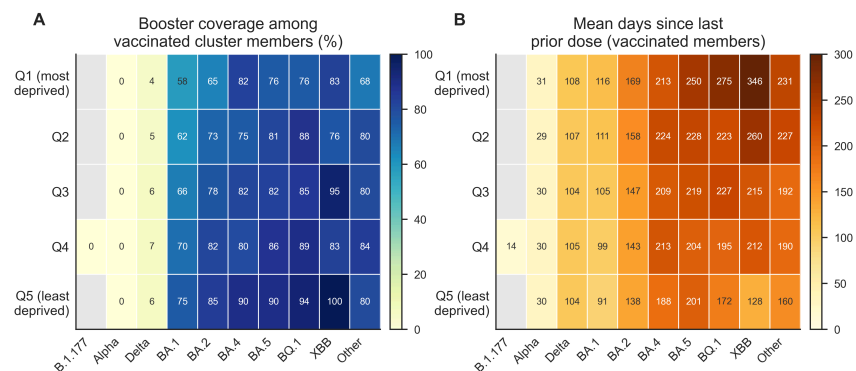
Part 2 Figure 1: Vaccinated cases over time



Part 2 Figure 2: Cluster vaccination by wave and category



Part 2 Figure 3: Vaccination-status mixing by wave



Part 2 Figure 6: Booster coverage and dose recency by SIMD

6.5. Key Tables

Table	File
Vaccination descriptives	vaccination_descriptives.csv
Vaccination cluster table	vaccination_cluster_table.csv
Cluster category summary	cluster_category_summary.csv
Vaccination cluster wave/category summary	vaccination_cluster_wave_category_summary.csv
SIMD-domain vaccination gradients	supp_simd_domain_vaccination_gradients.csv

7. Part 3: Policy Restrictions and Cluster Structure

7.1. Focus

Part 3 examines how Scottish COVID-19 restriction periods align with genomic cluster size, geographic dispersion, and demographic mixing. Because restriction periods are strongly confounded with calendar time, variant waves, immunity, and surveillance changes, the analysis is descriptive and associational.

7.2. Policy Period Codes

Policy intensity is an ordinal restriction score where higher values indicate more restrictive conditions. The genomic cluster analysis begins in P3, so E0, L1, P1, and P2 are included below for code completeness but are not observed in the Part 3 cluster models.

Code	Policy period label	Dates	Intensity	In cluster analysis
E0	Emergence	2020-03-01 to 2020-03-23	15	No
L1	First lockdown	2020-03-24 to 2020-05-28	100	No
P1	Route map phase 1	2020-05-29 to 2020-06-18	72	No
P2	Route map phase 2	2020-06-19 to 2020-07-09	52	No
P3	Route map phase 3	2020-07-10 to 2020-10-01	30	Yes
T1	Pre-tier tightening	2020-10-02 to 2020-11-01	55	Yes
F5	Five-tier framework	2020-11-02 to 2021-01-04	65	Yes
L2	Second lockdown	2021-01-05 to 2021-04-01	95	Yes
SL	Stay local - Level 3	2021-04-02 to 2021-04-25	65	Yes
L3	Level 3	2021-04-26 to 2021-05-16	55	Yes
L21	Level 2 / Level 1	2021-05-17 to 2021-07-18	38	Yes

Code	Policy period label	Dates	Intensity	In cluster analysis
L0	Level 0	2021-07-19 to 2021-08-08	20	Yes
NN	Near-normal	2021-08-09 to 2021-11-28	10	Yes
OM	Omicron wave	2021-11-29 to 2022-01-23	42	Yes
FE	Final easing	2022-01-24 to 2022-04-17	15	Yes
PR	Post-restriction	2022-04-18 to 2023-05-05	3	Yes

7.3. Key Results

Analysis	Result	Interpretation
Policy intensity and weekly median log cluster size	Spearman rho = 0.74	Strong positive correlation, likely driven by wave confounding.
Policy intensity and weekly log datazones	Spearman rho = 0.58	Moderate positive correlation, also confounded by wave and surveillance context.
Policy intensity and mean SIMD excess discordance	Spearman rho = 0.02	No meaningful correlation with SIMD mixing.
Policy intensity and mean age excess discordance	Spearman rho = 0.59	Positive correlation, but not interpretable causally.

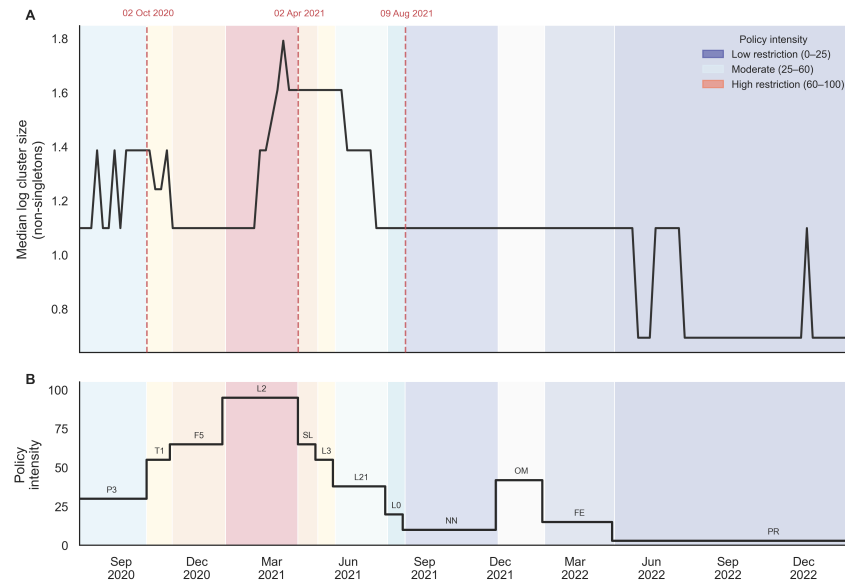
ITS transition	Cluster size	Geographic spread	Mixing metrics	Interpretation
T1 onset, October 2020	beta = -0.13, p = 0.22	beta = -0.07, p = 0.59	No significant changes	No detectable structural change at the tier introduction.
L2 to SL, April 2021	beta = -0.21, p = 0.034	beta = -0.36, p = 0.006	No significant changes	Decline likely reflects Alpha-wave tail dynamics rather than easing itself.
NN onset, August 2021	beta = +0.12, p = 0.068	beta = +0.32, p = 0.015	No significant changes	Geographic spread increased after legal distancing requirements ended, within strong Delta-wave context.

7.4. Main Insight

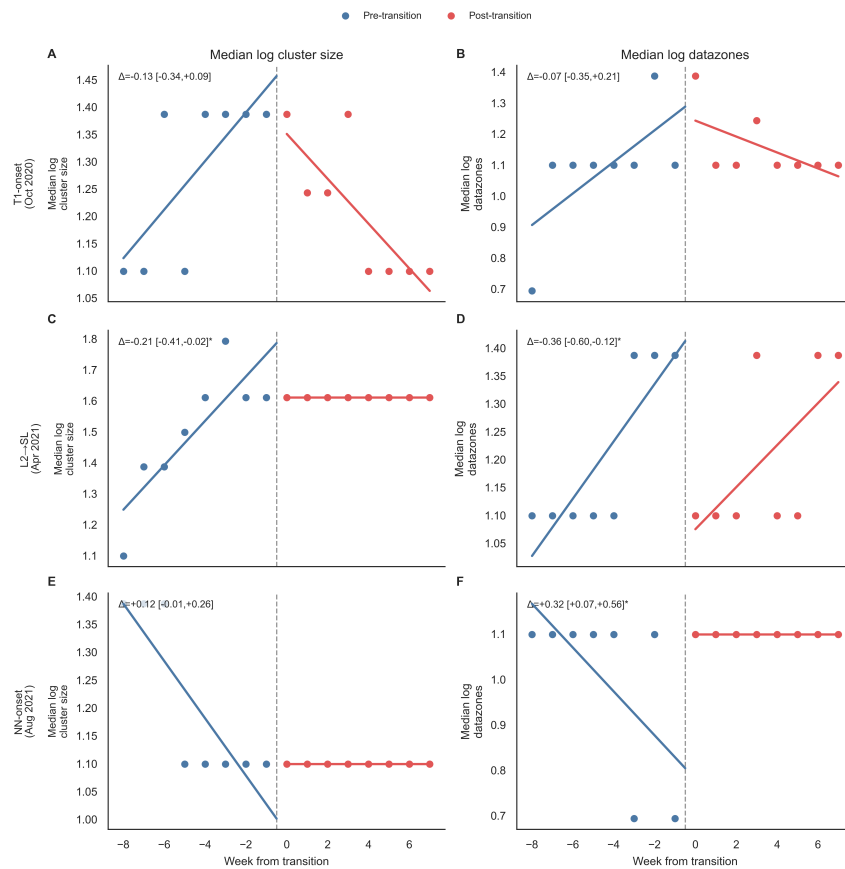
Restriction intensity was correlated with cluster size and spread across the whole epidemic, but these correlations are mainly contextual rather than causal. The strongest transition-level signal was increased geographic spread after the August 2021 move to near-normal restrictions. Demographic mixing did not change significantly across the selected policy transitions, supporting the Part 1 interpretation that cluster composition reflects local social structure more than broad policy intensity.

7.5. Key Figures

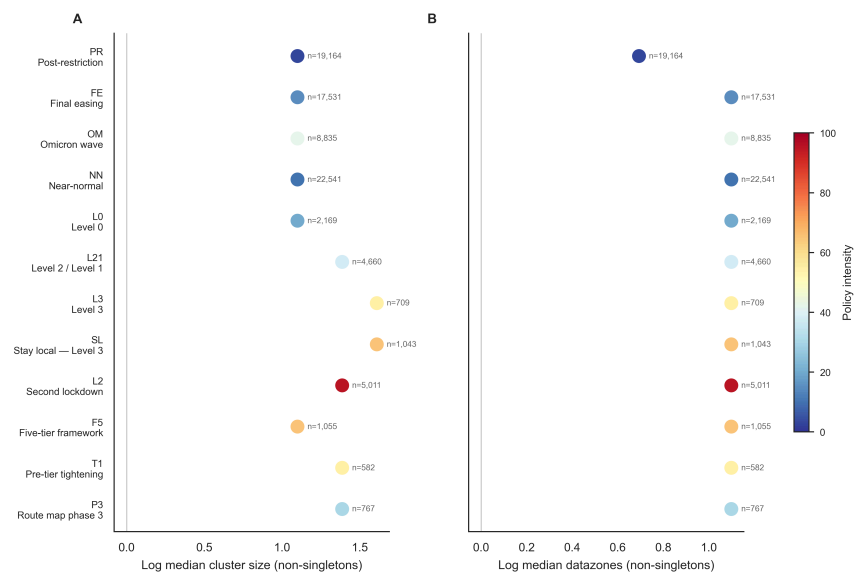
Figure	What it shows
Part 3 Figure 1	Weekly cluster outcomes shown against policy-period shading and restriction intensity.
Part 3 Figure 2	Interrupted time-series plots for the T1 onset, L2 to SL, and NN onset transitions.
Part 3 Figure 3	Median cluster size and geographic spread compared across observed policy periods.



Part 3 Figure 1: Weekly time series and policy context



Part 3 Figure 2: Interrupted time-series transitions



Part 3 Figure 3: Policy-period outcomes

7.6. Key Tables

Table	File
Policy-period descriptives	period_descriptives.csv
Intensity correlations	intensity_correlations.csv
ITS coefficients	its_coefficients.csv
Lagged intensity correlations	supp_lagged_intensity_correlations.csv
ITS window sensitivity	supp_its_window_sensitivity.csv

8. Overall Interpretation

The strongest recurring conclusion is that genomic clusters cannot be interpreted as direct measures of transmission scale without accounting for surveillance and epidemic context. Deprivation, vaccination, and policy context all matter, but they operate through different parts of the system:

- Deprivation is most visible in cluster composition and demographic mixing.
- Vaccination status mainly reflects rollout timing and background population coverage among sequenced cases.
- Policy periods help contextualise the epidemic timeline, but policy effects cannot be separated cleanly from variant waves and calendar time in these data.

9. Manuscript Advice Request

I would appreciate advice on the best manuscript strategy. My current view is that Part 1 is the strongest standalone paper because it has the clearest inferential structure and central argument: socioeconomic deprivation does not map simply onto larger or more geographically dispersed genomic clusters once surveillance and epidemic context are accounted for, but it is associated with demographic mixing and cluster composition.

One option is therefore to develop Part 1 as the main manuscript and use Parts 2 and 3 as thesis chapters or as contextual supporting material. Another option is to draw selectively across all three parts to write a broader paper on the social, vaccination, and policy context of SARS-CoV-2 genomic clusters in Scotland. My concern is that combining all three parts may make the manuscript too broad, while focusing only on Part 1 may

underuse the richer vaccination and policy analyses. I would be grateful for advice on which framing is most publishable and whether Parts 2 and 3 should be kept separate, used as supplementary context, or developed into future papers.